

# Day 10 - IPv4 Header

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## 10. THE IPv4 HEADER

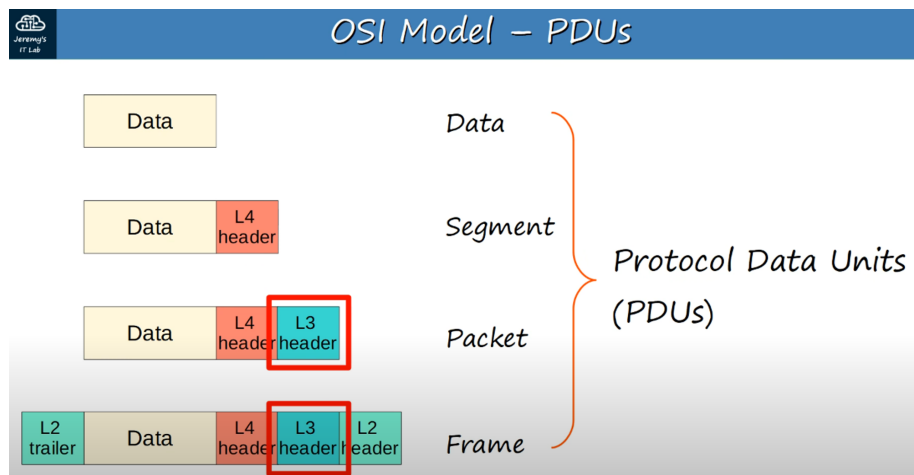
INTERNET PROTOCOL version 4 HEADER or IPv4 HEADER

HEADER is used at LAYER 3 to help send data between devices on separate networks, even on other sides of the world over the Internet.

This is known as ROUTING.

THE IPv4 HEADER is used to ENCAPSULATE a TCP or UDP Segment.

To Review:



FIELDS OF THE IPv4 HEADER

| Offsets | Octet | 0                      | 1        | 2               | 3            |
|---------|-------|------------------------|----------|-----------------|--------------|
| Octet   | Bit   | 0                      | 1        | 2               | 3            |
| 0       | 0     | Version                | IHL      | DSCP            | ECN          |
| 4       | 32    | Identification         | Flags    | Fragment Offset | Total Length |
| 8       | 64    | Time To Live           | Protocol | Header Checksum |              |
| 12      | 96    | Source IP Address      |          |                 |              |
| 16      | 128   | Destination IP Address |          |                 |              |
| 20      | 160   |                        |          |                 |              |
| 24      | 192   |                        |          |                 |              |
| 28      | 224   |                        |          |                 |              |
| 32      | 256   |                        |          |                 |              |

| FIELD           | # OF BITS |
|-----------------|-----------|
| VERSION         | 4         |
| IHL             | 4         |
| DSCP            | 6         |
| ECN             | 2         |
| TOTAL LENGTH    | 16        |
| IDENTIFICATION  | 16        |
| FLAGS           | 3         |
| FRAGMENT OFFSET | 13        |
| TIME TO LIVE    | 8         |

|                     |         |
|---------------------|---------|
| PROTOCOL            | 8       |
| HEADER CHECKSUM     | 16      |
| SOURCE ADDRESS      | 32      |
| DESTINATION ADDRESS | 32      |
| OPTIONS             | 320 Max |

#### VERSION:

- LENGTH is 4 bits.
- IDs version of IP used (IPv4 or IPv6)
  - **IPv4 = 0100 in Binary (Decimal 4)**
  - **IPv6 = 0110 in Binary (Decimal 6)**

#### INTERNET HEADER LENGTH (IHL):

- LENGTH is 4 bits.
- Final field of IPv4 Header (Options) is variable in length so this field is necessary to indicate the total length of the header.
- IDs the length of the header in 4-BYTE INCREMENTS.
- The MINIMUM value is 5 (5 \* 4-bytes = 20 bytes) - Empty OPTIONS Field
- The MAXIMUM value is 15 (15 \* 4-bytes = 60 bytes)

MINIMUM IPv4 HEADER LENGTH = 20 Bytes! MAXIMUM IPv4 HEADER LENGTH = 60 Bytes!

#### DSCP (Differentiated Services Code Point):

- LENGTH is 6 bits.
- Used for QoS (Quality of Service)
- Used to prioritize delay-sensitive data (streaming voice, video, etc.)

#### ECN (Explicit Congestion Notification):

- LENGTH is 2 bits.
- Provides end-to-end (between two endpoints) notification of network congestion WITHOUT dropping packets.
- Optional feature that requires both endpoints, as well as the underlying network infrastructure to support it.

#### TOTAL LENGTH:

- LENGTH is 16 bits.
- Indicates the TOTAL length of the packet (L3 Header + L4 Segment)
- Measured in bytes (not 4-byte increments like IHL)
- Minimum value of 20 Bytes (IPv4 Header with NO encapsulated data)
- Maximum value of 65,535 (MAXIMUM 16-bit value) =  $2^{16}$

#### IDENTIFICATION:

- LENGTH is 16 bits.
- If a packet is fragmented due to being too large, this field is used to identify which packet the fragment belongs to.
- All fragments of the same packet will have their own IPv4 header with the same value in this field.
- Packets are fragmented, if larger than the MTU (Maximum Transmission Unit)
- The MTU is usually 1500 bytes (Max size of an Ethernet frame)
- Fragments are reassembled by the receiving host.

#### FLAGS:

- LENGTH is 3 bits
- Used to control/identify fragments.
- Bit 0: Reserved, always set to 0.
- Bit 1: Don't Fragment (DF bit), used to indicate a packet that should not be fragmented.
- Bit 2: More Fragments (MF bit), set to 1 if there are more fragments in the packet, set to 0 for the last fragment or NO fragments.

#### FRAGMENT OFFSET:

- LENGTH is 13 bits
- Used to indicate the position of the fragment within the original, unfragmented IP Packet.
- Allows fragmented packets to be reassembled even if the fragments arrive out of order.

#### TIME TO LIVE (TTL):

- LENGTH is 8 bits
- A router will drop a packet with a TTL of 0
- Used to prevent infinite loops
- Originally designed to indicate a packet's maximum lifetime in seconds.
- In practice, indicates a 'hop count': each time the packet arrives at a router, the router decreases the TTL by 1.
- Recommended default TTL is 64.

#### PROTOCOL:

- LENGTH is 8 bits
- Indicates the protocol of the encapsulated Layer 4 PDU
- Value of 1 : ICMP
- Value of 6 : TCP
- Value of 17 : UDP
- Value of 89 : OSPF (Dynamic Routing Protocol)
- List of protocol numbers on Wikipedia : [List of IP Protocol Numbers](#)

#### HEADER CHECKSUM:

- LENGTH is 16 bits
- A calculated checksum used to check for errors in the IPv4 header.
- When a router receives a packet, it calculates the checksum of the header and compares it to the one in this field of a header.
- If they do not match, the router drops the packet.
- Used to check for ERRORS only in the IPv4 Header.
- IP relies on the encapsulated protocol to detect errors in the encapsulated data.
- Both TCP and UDP have their own checksum fields to detect errors in the encapsulated data.

#### SOURCE and DESTINATION:

- LENGTH is 32 bits each
- SOURCE IP = IPv4 ADDRESS of the Sender of the Packet.
- DESTINATION IP = IPv4 ADDRESS of the intended Receiver of the Packet.

#### OPTIONS:

- LENGTH is 0-320 bits
- Optional / Rarely Used
- If the IHL field is greater than 5, it means that Options are present.